Grover's Algorithm

5/6 topics completed

Gain intuition on how Grover's search algorithm works and code it end-to-end.



Introduction to Grover Search

Become familiar with the basic components of Grover's Algorithm.



Searching with Geometry

Study Grover's Algorithm from a geometric perspective.



Searching with Circuits

Learn the relation between phase kickback and Grover search.



Steps and Speedups

Compute the optimal number of Grover steps.



Multiple Solutions

Implement Grover to search for multiple solutions.



Amplitude Amplification

Implement Amplitude Amplification to non-uniform states.



A Mixed Bag

Analyze the QPE output for an arbitrary input state.